

Chapter 1

Dual Vector Spaces

Definition 1.1. Let V be a vector space over F . The dual space, denoted V^* , is the vector space of all linear maps from V to F :

$$V^* := \{T : V \rightarrow F \mid T \text{ is linear}\}$$

Remark. A vector $v \in V$ is an algebraic object (a geometric arrow, a polynomial, a matrix). A dual vector $f \in V^*$ is a linear scalar-valued function. It is best understood as an abstract “measurement object” that takes a vector and returns a scalar value in the base field F , representing some linear property of that vector.

If $V = F^n$ consists of column vectors, V^* consists of row vectors. The evaluation $f(v)$ is exactly the matrix multiplication of a $1 \times n$ row vector with an $n \times 1$ column vector to yield a 1×1 scalar, i.e. the dot product (we will see what this means geometrically in $\mathbb{R}^n$ later).

Example 1.2. Let $V = \mathbb{R}[x]_{\leq 2}$ be the vector space of polynomials in $\mathbb{R}$ of degree at most 2. The elements are polynomials $p(x) = a + bx + cx^2$. We can define several distinct dual vectors (“measurement objects”) in V^* :

- Pointwise evaluation: Let $E_2 \in V^*$ be defined by $E_2(p) = p(2)$. This function takes a polynomial and returns its value at $x = 2$. It is linear, so $E_2 \in V^*$.
- Differentiation: Let $D_0 \in V^*$ be defined by $D_0(p) = p'(0)$. This measures the slope of the polynomial at the origin (which evaluates to the coefficient b).
- Integration: Let $I \in V^*$ be defined by $I(p) = \int_0^1 p(x)dx$. This measures the area under the curve.

Each of these is a distinct vector in the dual space V^* .

The vector space structure on V^* is given by pointwise addition and scalar multiplication. For $f, g \in V^*$ and $\lambda \in F$:

$$\begin{aligned}(f + g)(v) &= f(v) + g(v) \\ (\lambda f)(v) &= \lambda(f(v))\end{aligned}$$

This space is also denoted as $\text{Hom}(V, F)$. More generally, $\text{Hom}(V, W)$ denotes the vector space of linear maps from V to W .

Exercise. Verify the vector space axioms for $\text{Hom}(V, W) := \{T : V \rightarrow W \mid T \text{ is linear}\}$, and hence also for V^* .

Example 1.3. Consider $V = F^n$, viewed as column vectors. All linear maps $V \rightarrow F \cong F^1$ are given by multiplication by matrices, in this case row vectors $u \in M_{1 \times n}(F)$. We can identify the dual with row vectors $M_{1 \times n}(F)$, so that if $u \in M_{1 \times n}(F)$ and $v \in M_{n \times 1}(F)$, we get the linear transformation $T_u : M_{n \times 1}(F) \rightarrow M_{1 \times 1}(F) \cong F$. Precisely, this gives an isomorphism $M_{n \times 1}(F) \rightarrow M_{1 \times n}(F)^*$, $u \mapsto T_u$.

The preceding example shows that dualising a finite-dimensional vector space produces another vector space of the same dimension ($\dim V = n \implies V \cong F^n \implies V^* \cong (F^n)^*$).

1.1 Dual Transformations

In addition to dualising vector spaces, we can also dualise maps.

Definition 1.4. Given a linear transformation $T : V \rightarrow W$, the dual transformation $T^* : W^* \rightarrow V^*$ is defined as:

$$T^*(f) := f \circ T$$

Remark. Given a linear map $T : V \rightarrow W$, the dual map $T^* : W^* \rightarrow V^*$ reverses the direction. When translated into matrices via bases, this exact operation yields the transpose: ${}_{\mathcal{B}^V}[T^*]_{\mathcal{C}^W} = ({}_{\mathcal{C}}[T]_{\mathcal{B}})^{\top}$, which is the “dual” idea (we will see why this is the case later).

More specifically, if you want to evaluate an object $v \in V$ using a measuring object $f \in W^*$ after a transformation T has been applied, there are two “dual” perspectives. You can either push $v \in V$ forward via T , and then measure it with f . This is $f(T(v))$. Alternatively, the “dual” perspective is that you can pull f backwards via T^* , and measure the original $v \in V$ using a measurement object in V^* , namely $T^*(f)$. Formally, we say that T^* is the pullback induced by T .

Proposition 1.5. Given a linear transformation T , the dual T^* is also a linear transformation.

Proof. We have $T^*(f + g) = (f + g) \circ T = (f \circ T) + (g \circ T) = T^*f + T^*g$, and also $T^*(\lambda f) = (\lambda f) \circ T = \lambda(f \circ T) = \lambda T^*(f)$. $\square$

Example 1.6. Let $T = T_A : F^n \rightarrow F^m$. Identifying $F^n \cong M_{n \times 1}(F)$ and $(F^n)^* \cong M_{1 \times n}(F)$, we have $T_A^*(u)v = u(Av)$, which implies $T_A^*u = uA$. Thus, $T_A^* : M_{1 \times m}(F) \rightarrow M_{1 \times n}(F)$ is the operation of right multiplication by A on row vectors.

Proposition 1.7. Given a composition $U \xrightarrow{S} V \xrightarrow{T} W$, we have the corresponding dual maps $U^* \xleftarrow{S^*} V^* \xleftarrow{T^*} W^*$, and:

$$(T \circ S)^* = S^* \circ T^*$$

Also, $I_V^* = I_{V^*}$.

Proof. The first assertion is associativity of composition: $(T \circ S)^*(f) = f \circ (T \circ S) = (f \circ T) \circ S = S^*(T^*(f))$. The second follows from the identity for composition: $f \circ I_V = f$. $\square$

Corollary 1.8. *If $T : V \rightarrow W$ is invertible, so is T^* , and $(T^{-1})^* = (T^*)^{-1}$.*

Proof. We have $T^* \circ (T^{-1})^* = (T^{-1} \circ T)^* = I_V^* = I_{V^*}$. Similarly $(T^{-1})^* \circ T^* = I_W^* = I_{W^*}$. $\square$

1.2 Dual Basis

Definition 1.9. Given a finite basis $(v_1, \dots, v_n)$ of a vector space V over F , let $v_1^\vee, \dots, v_n^\vee \in V^*$ be the functions defined by:

$$v_i^\vee(\lambda_1 v_1 + \dots + \lambda_n v_n) = \lambda_i$$

In particular, we have the formula:

$$v_i^\vee(v_j) = \delta_{ij} := \begin{cases} 1, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases}$$

Proposition 1.10. *The n -tuple $(v_1^\vee, \dots, v_n^\vee)$ of elements of V^* is a basis.*

Proof. First we show that the tuple is linearly independent. If $f := \lambda_1 v_1^\vee + \dots + \lambda_n v_n^\vee = 0$, then evaluating at v_i gives $f(v_i) = \lambda_i = 0$ for all i .

Next we show that it spans. If $f \in V^*$, we claim that:

$$f = f(v_1)v_1^\vee + \dots + f(v_n)v_n^\vee \tag{1.1}$$

Let g be the right-hand side. Then $g(v_i) = f(v_i)$ for all i , by definition. By linearity and the fact that $(v_1, \dots, v_n)$ span V , we get $g = f$. $\square$

Definition 1.11. The basis $\mathcal{B}^\vee := (v_1^\vee, \dots, v_n^\vee)$ is called the *dual basis* of $\mathcal{B} = (v_1, \dots, v_n)$.

This yields the companion formula for all $v \in V$:

$$v = v_1^\vee(v)v_1 + \dots + v_n^\vee(v)v_n.$$

Corollary 1.12. *For V a finite-dimensional vector space, V^* is also finite-dimensional, and*

$$\dim V^* = \dim V$$

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This together with the injectivity of ι yields:

Corollary 1.13. *For V a finite-dimensional vector space, $\iota : V \rightarrow (V^*)^*$ is an isomorphism.*

Proof. We have seen ι is linear and injective. Since V is finite-dimensional, so is $(V^*)^*$, and it has the same dimension. Now ι is a linear map between finite-dimensional vector spaces of the same dimension; so it is an isomorphism. $\square$

1.2.1 The Double Dual

Given a vector space V , we define the map:

$$\iota : V \rightarrow (V^*)^*, \quad v \mapsto \iota_v, \quad \text{where } \iota_v(f) = f(v)$$

Remark. Note that no basis is required in order to define this map.

Proposition 1.14. *The map ι is injective and linear (assuming V is finite-dimensional).*

Proof. Linearity follows because $\iota_{v+v'}(f) = f(v+v') = f(v) + f(v') = \iota_v(f) + \iota_{v'}(f)$, and similarly $\iota_{\lambda v}(f) = \lambda \iota_v(f)$.

For injectivity, suppose $v \in V$ is nonzero. Then we have an inclusion $F \cdot v = \text{Span}(v) \subseteq V$ of subspaces, so the restriction map $V^* \rightarrow (F \cdot v)^*$ is surjective. That is, there exists $f \in V^*$ with $f(v) \neq 0$. Then $\iota_v(f) = f(v) \neq 0$ as well, so $\iota_v \neq 0$. Thus, $\ker(\iota) = \{0\}$, so ι is injective. $\square$

We will see that ι is an isomorphism when V is finite-dimensional. To do this, we find a basis for V^* .

Given a finite-dimensional vector space V and a basis $\mathcal{B} \subseteq V$, we form the bases:

$$\mathcal{B}^\vee \subseteq V^*, \quad \mathcal{B}^{\vee\vee} \subseteq V^{**}$$

We also have the isomorphism $\iota : V \xrightarrow{\sim} (V^*)^*$.

Proposition 1.15. *Taking the double dual basis recovers the image under the isomorphism ι :*

$$\mathcal{B}^{\vee\vee} = \iota(\mathcal{B})$$

Proof. Let $\mathcal{B} = (v_1, \dots, v_n)$, $\mathcal{B}^\vee = (v_1^\vee, \dots, v_n^\vee)$, and $\mathcal{B}^{\vee\vee} = (v_1^{\vee\vee}, \dots, v_n^{\vee\vee})$. Then we have:

$$\iota(v_i)(v_j^\vee) = v_j^\vee(v_i) = \delta_{ij} = v_i^{\vee\vee}(v_j^\vee)$$

Fix i . Since $\iota(v_i)(v_j^\vee) = v_i^{\vee\vee}(v_j^\vee)$ for all j , by taking linear combinations of the basis $(v_1^\vee, \dots, v_n^\vee)$ of V^* , we get $\iota(v_i)(f) = v_i^{\vee\vee}(f)$ for all $f \in V^*$. That is, $\iota(v_i) = v_i^{\vee\vee}$. $\square$

As a consequence, $v_i^{\vee\vee}$ depends only on v_i , not on v_j for $j \neq i$. This is not true about the dual basis: $v_i^\vee$ depends on all v_j . It is defined so that $v_i^\vee(v_j) = 0$ when $i \neq j$.

Example 1.16. Let $V = F^2$, with basis $\mathcal{B} = (v_1, v_2)$ for $v_1 = e_1$ and $v_2 = \lambda e_1 + e_2$. Then $v_1^\vee = e_1^\vee - \lambda e_2^\vee$. We see that it depends on λ , hence on v_2 . In terms of column and row vectors: $\mathcal{B} = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} \lambda \\ 1 \end{pmatrix} \right)$ and $\mathcal{B}^\vee = ((1 \ 0), (-\lambda \ 1))$.

1.2.2 Annihilators

Definition 1.17. Given $W \subseteq V$, we define $W^\perp \subseteq V^*$ by:

$$W^\perp := \{f \in V^* \mid f(w) = 0, \forall w \in W\}$$

In other words, it is the set of vectors which annihilate all of W .

Example 1.18. Suppose V has basis $\mathcal{B} = (v_1, \dots, v_n)$ and $W = \text{Span}(v_1, \dots, v_k)$ for some $1 \leq k \leq n$. Then $W^\perp = \text{Span}(v_{k+1}^\vee, \dots, v_n^\vee)$. For example, if $V = M_{n \times 1}(F)$ and $v_i = e_i$ is the standard basis, then

$$W = \left\{ \begin{pmatrix} a_1 \\ \vdots \\ a_m \\ 0 \\ \vdots \\ 0 \end{pmatrix} : a_1, \dots, a_m \in F \right\}, \quad W^\perp = \{(0 \quad \dots \quad 0 \quad b_1 \quad \dots \quad b_{n-m}) : b_1, \dots, b_{n-m} \in F\}$$

Proposition 1.19. If V is finite-dimensional, and $W \subseteq V$ a subspace, then:

$$\dim V = \dim W + \dim W^\perp$$

Remark. When inner products are introduced, they provide an isomorphism $V \rightarrow V^*$, and using this, the preimage of $W^\perp$ is the usual subspace of vectors which dot with all $w \in W$ to zero.

1.3 Dualisation and Transpose

Theorem 1.20. Let V and W be finite-dimensional vector spaces with bases $\mathcal{B}$ and $\mathcal{C}$, respectively. Let $T : V \rightarrow W$ be a linear transformation. Then:

$${}_{\mathcal{B}^\vee}[T^*]_{\mathcal{C}^\vee} = ({}_c[T]_{\mathcal{B}})^\top$$

Proof. Let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$, with dual bases $\mathcal{B}^\vee = (v_1^\vee, \dots, v_n^\vee)$ and $\mathcal{C}^\vee = (w_1^\vee, \dots, w_m^\vee)$. Then

$$T^*(w_j^\vee)(v_i) = w_j^\vee \circ T(v_i) = w_j^\vee(T(v_i))$$

By definition, the left-hand side is the ij -th entry of the matrix ${}_{\mathcal{B}^\vee}[T^*]_{\mathcal{C}^\vee}$, whereas the right-hand side is the ji -th entry of the matrix ${}_c[T]_{\mathcal{B}}$. $\square$

Example 1.21. In the case $V = F^n, W = F^m$, we can verify the theorem directly. Every linear transformation from V to W is of the form T_A for $A \in M_{m \times n}(F)$. We know the formula ${}_{(e_1, \dots, e_m)}[T_A]_{(e_1, \dots, e_n)} = A$. If we identify F^n with column vectors, and $(F^n)^*$ with row vectors, then for $u \in M_{1 \times n}(F)$ a row vector, thought of as an element of V^* , we have

$$[u]_{(e_1^\vee, \dots, e_n^\vee)} = u^\top$$

We found previously that $T_A^*(T_u) = T_{uA}$, left multiplication by the row vector $uA \in M_{1 \times n}(F) \cong V^*$. We then get:

$$[T_A^*u]_{(e_1^\vee, \dots, e_n^\vee)} = [uA]_{(e_1^\vee, \dots, e_n^\vee)} = (uA)^\top = A^\top u^\top = A^\top [u]_{(e_1^\vee, \dots, e_m^\vee)}$$

This implies ${}_{\mathcal{B}^\vee}[T_A^*]_{\mathcal{C}^\vee} = A^\top$, by the definition of the matrix of a linear transformation.

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